

Customer Shopping Behavior Analysis

Insights from 3,900 purchases to guide strategy





Dataset Overview

Total Purchases

3,900

Data Columns

18 (demographics & behavior)

Missing Values

37 (only Review Rating)

Data Preparation

01

Load

Imported with pandas

02

Explore

Structure check, summary stats

03

Impute

Review Rating → median

04

Engineer

Age groups, purchase frequency

05

Integrate

PostgreSQL connection

Revenue by Gender

Female customers generate slightly higher total revenue than male customers

Female

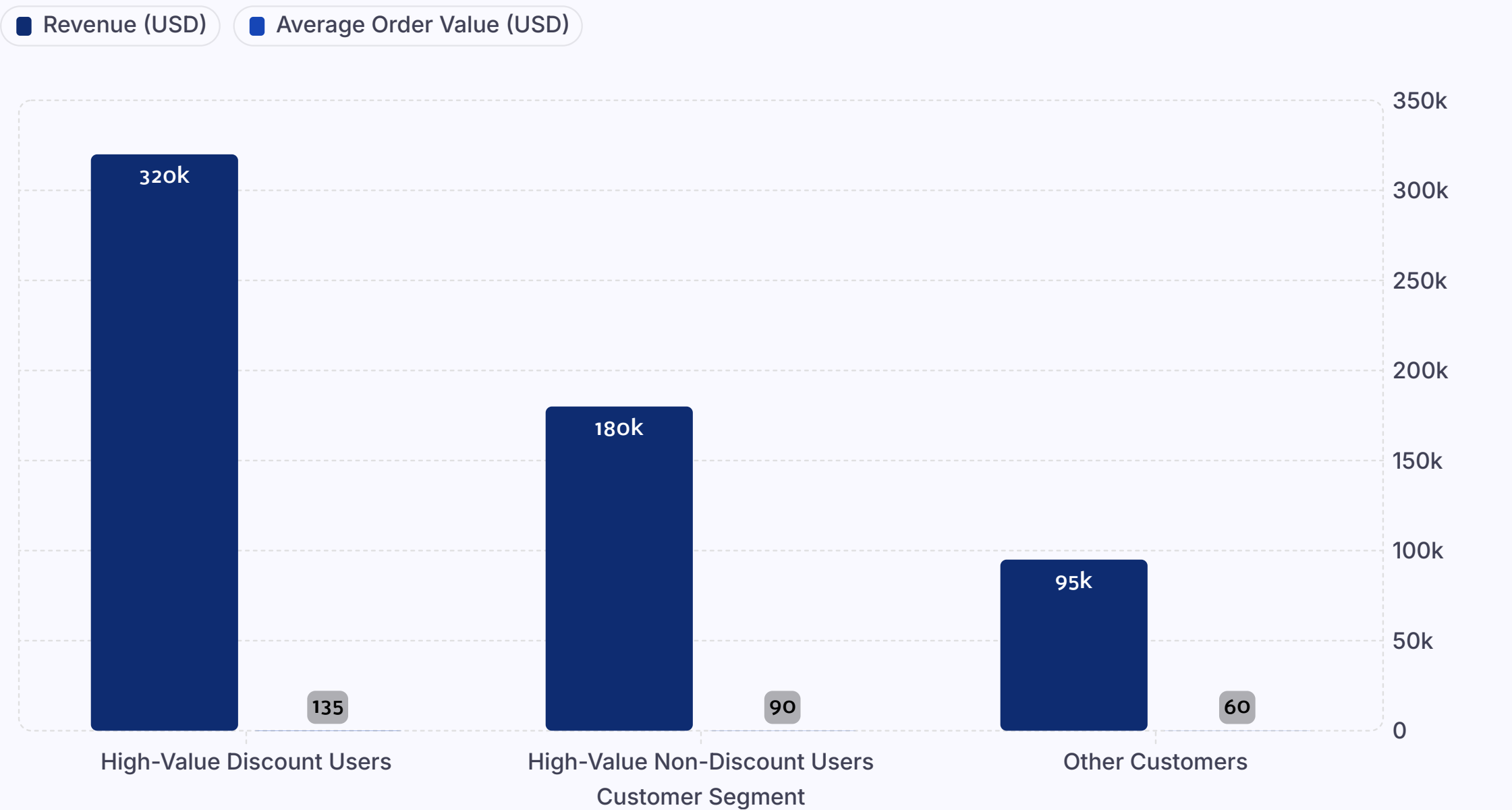
Slightly more than 50%

Male

Slightly less than 50%

High-Value Discount Users

A significant portion of our highest-spending customers are actively using discounts, driving substantial revenue.



This chart illustrates that "High-Value Discount Users" contribute the largest share of revenue and have a significantly higher Average Order Value (AOV) compared to both non-discount high-value customers and other customer segments. This suggests that strategic discounts are effectively incentivizing increased spending among our most valuable customers.

High-Value Discount Users



High spenders using discounts – "Smart Shoppers"

- ❑ Target premium customers with exclusive offers

Top-Rated Products



Blouse

5★ – Highest satisfaction



Dress

5★ – Consistent excellence



Shirt

4★ – Strong approval

Shipping & Spend



Express

\$65 average – +12% per transaction

Standard

\$58 average

Subscription Impact



Higher Spend

68%

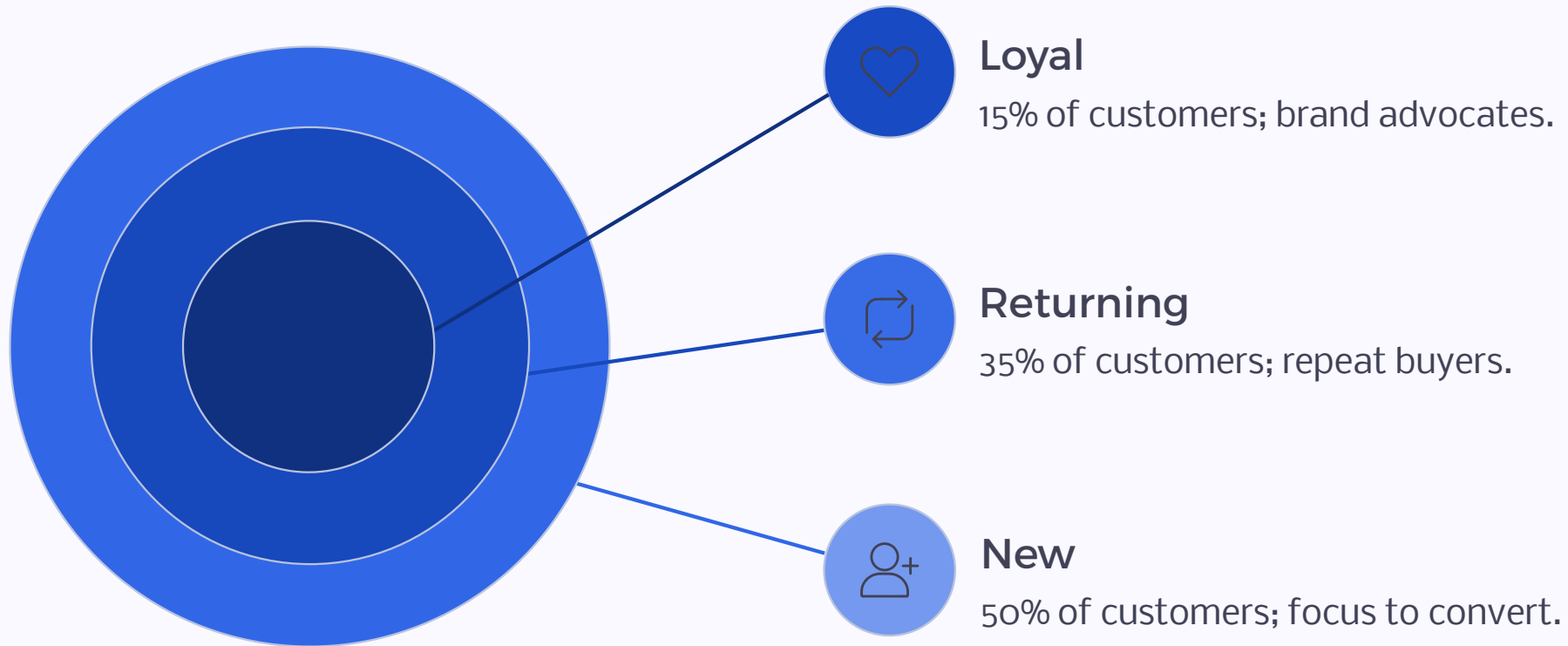
Revenue Share

45% from subscribers

Loyalty Rate

78% repeat purchases

Customer Segmentation



Convert New → Returning → Loyal

Strategic Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to raise retention



Targeted Marketing

Focus on high-revenue segments & express users



Product Positioning

Highlight top-rated products in campaigns



